SoccerGMR Ablation Study: Stage 2 G1 Selection vs. mAP Selection (8 Backbones)

Comparing GMR Baseline vs. GMR + DQ-CGPv3 vs. GEQ with Val G1 Selection vs. GEQ with Val mAP Selection on Standard Test (1,036 queries) and Full-Decodable Test (2,970 queries).

		Standard Test Split (1,036 queries)				Full-Decodable Test Split (2,970 queries)			
Backbone	**Configuration**	**mAP (%)**	**G1-AUC (%)**	**Rej-F1@.5 (%)**	**AUROC (%)**	**mAP (%)**	**G1-AUC (%)**	**Rej-F1@.5 (%)**	**AUROC (%)**
Moment-DETR	1. GMR Baseline	6.68	29.04	67.94	70.20	5.47	27.01	72.08	78.91
	2. GMR + DQ-CGPv3	10.66	34.85	72.20	72.09	9.99	32.82	74.92	80.24
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	**12.64**	**42.92**	**75.48**	73.48	**11.63**	**40.55**	**78.93**	80.37
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	12.63	35.51	68.77	**75.17**	11.56	33.45	69.83	**81.94**
EaTR	1. GMR Baseline	16.49	37.90	65.60	76.56	15.19	36.53	67.38	81.59
	2. GMR + DQ-CGPv3	16.02	35.82	65.27	**77.19**	15.35	34.99	65.29	80.38
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	17.48	**44.39**	**75.43**	76.17	15.67	**42.50**	**78.73**	**83.04**
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	**17.49**	37.46	70.12	76.28	**15.98**	35.73	72.31	82.28
QD-DETR	1. GMR Baseline	14.62	38.72	71.48	**77.01**	12.65	36.93	72.49	81.75
	2. GMR + DQ-CGPv3	15.82	39.54	69.02	74.24	13.93	38.05	71.63	81.64
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	**17.28**	**42.35**	**72.25**	75.18	**14.20**	**40.24**	**75.61**	**82.48**
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	16.80	41.02	68.29	74.73	13.61	39.30	71.15	82.20
CG-DETR	1. GMR Baseline	**18.65**	34.50	57.47	**76.40**	**14.67**	33.53	63.23	81.81
	2. GMR + DQ-CGPv3	13.49	36.84	75.04	74.63	12.49	35.40	76.97	79.33
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	14.75	**43.76**	**75.31**	75.77	13.17	**42.05**	**78.87**	**82.88**
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	14.75	33.30	58.42	75.41	13.62	33.01	62.98	82.44
BM-DETR	1. GMR Baseline	4.00	36.32	**75.00**	71.55	3.65	33.93	78.58	79.11
	2. GMR + DQ-CGPv3	7.74	38.02	74.43	70.69	7.32	36.15	77.84	79.64
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	7.82	**43.35**	**75.00**	72.09	7.32	**40.89**	**78.80**	79.15
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	**8.69**	34.85	71.33	**73.71**	**7.82**	32.88	75.15	**80.15**
BAM-DETR	1. GMR Baseline	**23.25**	42.92	70.43	**78.66**	20.62	40.75	71.79	80.99
	2. GMR + DQ-CGPv3	21.52	41.60	71.83	77.80	19.57	40.14	72.01	80.13
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	22.84	**43.82**	**74.07**	77.41	20.74	**41.76**	**77.58**	**83.32**
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	22.83	39.28	65.46	77.08	**21.21**	37.94	68.07	82.87
Sim-DETR	1. GMR Baseline	20.73	37.57	74.69	72.97	19.18	35.68	75.72	78.87
	2. GMR + DQ-CGPv3	20.63	38.90	75.43	75.84	19.62	37.24	75.69	80.20
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	21.43	**44.96**	**75.56**	**76.79**	19.50	**43.05**	**76.12**	**82.41**
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	**21.71**	36.79	67.73	76.70	**20.19**	35.58	70.09	80.95
UVCOM	1. GMR Baseline	7.12	35.95	**75.00**	70.03	6.40	34.32	**76.80**	**78.98**
	2. GMR + DQ-CGPv3	17.06	38.75	70.06	**76.83**	16.41	37.37	67.20	75.32
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	**18.40**	**42.62**	72.28	76.65	**17.12**	**40.80**	71.45	76.29
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	17.16	36.99	59.80	75.83	16.50	35.85	59.55	75.48

Method 3 (G1 Sel): Global Event Query fine-tuned with model selection on Val Mean G-mIoU@1 (0.4/0.5/0.6) (Official).
Method 4 (mAP Sel): Global Event Query fine-tuned with model selection on highest Val mAP.
G1-AUC (%): Area Under the Curve of G-mIoU@1 across tau in [0.00, 1.00]. Bold values: Best within backbone family.